Five-Minute Research Presentation

Bank Branch Network Contraction and Employer Startup Activity in North Carolina Counties, 2014-2023

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Slide 1: Title and one-sentence result (0:00-0:25)

On screen

- Bank Branch Network Contraction and Employer Startup Activity in North Carolina Counties, 2014-2023
- Public panel: 100 counties, 10 outcome years
- Result: substantial branch decline, but no precise, temporally stable county-level association with employer startups

Speaker notes

"I studied whether changes in North Carolina's full-service bank-branch network were related to the formation of genuinely new employer businesses. I built a reproducible county panel from five federal sources. The branch decline is clear; the estimated startup relationship is not. My internally frozen result is negative and imprecise; a clean-preperiod audit model is positive and imprecise. Both are observational."

Slide 2: Why the question matters (0:25-0:55)

On screen

- Young firms may rely on relationship lending and local information.
- Digital banking may substitute for some branch functions.
- Rural losses may be more consequential when alternatives are distant.

Speaker notes

"A branch can matter because a banker can collect information that a standardized application misses, especially for a young firm. But technology, fintech, and nearby branches across county lines can weaken that channel. So I did not assume fewer offices automatically means less access. I tested a narrower question about branch density and employer startups."

Slide 3: Research question and contribution (0:55-1:25)

On screen

> How is full-service bank-branch density associated with employer-startup activity after county and year fixed effects?

- North Carolina-specific
- Fully public and reproducible
- Firm-age-zero startups, not every establishment opening
- Association, not causation

Speaker notes

"My contribution is measurement and transparency, not discovering the mechanism first. Stronger national studies use restricted microdata and merger-based instruments. I use public county data and make a cleaner startup distinction: firm-age-zero establishments enter with a genuinely new employer firm, while the official entry rate also includes an existing firm opening another location."

Slide 4: Data and construction (1:25-2:05)

On screen

```
Startup rate = 1,000 × age-zero establishments(t)
                     / all establishments(t-1)

Branch density = 10,000 × full-service branches(t-1)
                        / county population(t-1)
```

- Census BDS, FDIC SOD, BEA, BLS LAUS, USDA RUCC
- 1,000 candidate county-years; five BDS `D` cells leave 995 primary observations

Speaker notes

"The denominator is lagged so same-year entrants do not enlarge their own benchmark. I count only FDIC SOD service codes 11 and 12. Five small Census startup cells are disclosure-suppressed, so I keep them missing rather than calling them zero. A timing audit found that the frozen June snapshot falls inside the BDS March-to-March outcome interval; I therefore also report a clean-preperiod model using the earlier June snapshot."

Slide 5: Empirical method (2:05-2:35)

On screen

```
StartupRate_it = county FE + year FE
               + beta × BranchDensity_i,t-1
               + lagged unemployment, log population, log PCPI
               + error_it
```

- County-clustered standard errors
- Rural interaction reports metro slope, nonmetro slope, and difference

Speaker notes

"County fixed effects compare each county with itself and remove stable differences. Year effects remove shocks common across the state. Clustering by county allows errors to remain correlated within a county over time. This still does not solve reverse causality: banks may close where they expect future activity to weaken."

Slide 6: Main evidence (2:35-3:30)

On screen

- Full-service branches: 2,554 in 2013 to 1,998 in 2022 (**−21.8%**)
- Headline slope: **−1.823**
- Clustered SE: 1.809
- 95% CI: [**−5.412, 1.767**]; N = 995
- One within-county SD corresponds to −0.745 startups per 1,000, but remains uncertain
- Clean-preperiod slope: **2.028**; 95% CI: [**−1.383, 5.439**]; N = 895

Speaker notes

"The branch contraction is not ambiguous. The regression is. The frozen model estimates minus 1.823, with an interval from minus 5.412 to plus 1.767. But its June snapshot overlaps the outcome interval. Moving the exposure back to a clean preperiod estimates plus 2.028, with an interval from minus 1.383 to plus 5.439. Both are imprecise. The correct conclusion is no precise or temporally stable sign. The data do not reveal that branches reduce startups or that branches do not matter."

Slide 7: Robustness, limitations, and the inconvenient result (3:30-4:25)

On screen

- COVID exclusion, population weighting, first differences, secondary outcomes: all imprecise
- Metro and nonmetro slopes: both imprecise
- Branch-count exposure: approximately zero
- Clean-preperiod exposure: positive but imprecise; sign reverses
- Poisson count model: negative and statistically distinguishable; retained as secondary
- Density is not distance/access; BDS excludes nonemployers; closures are endogenous

Speaker notes

"Most checks remain inconclusive. The clean-preperiod sign reversal shows that timing matters. A Poisson count model gives a negative interval excluding zero, so I report it rather than hide it. I do not promote it because it changes the functional form, shares the same endogeneity, and is one result in a correlated sequence. My biggest measurement limitation is that branches per capita is not travel distance or credit access. My biggest identification limitation is endogenous closure placement."

Slide 8: Implication and next step (4:25-5:00)

On screen

- Do not infer local access from statewide branch counts alone.
- Monitor exact location, travel time, cross-county alternatives, digital access, and nonbank providers.
- Next study: geocoded closures + sub-county outcomes + merger-related variation.

Speaker notes

"My policy conclusion is about better measurement, not opening a branch because one regression says so. North Carolina should identify where an office loss actually changes travel time and small-business options. The credible research next step is a geocoded, sub-county design using exact closure dates and merger-related variation. Every result, including the inconclusive and mixed findings, is traceable and reproducible."

Likely questions after the presentation

1. **Why not claim causality?** Fixed effects do not remove time-varying local shocks or banks' expectations.
2. **Why not use all business applications?** Applications are intentions and include many nonemployers; BDS age zero measures operating employer-firm entry.
3. **Why did the Poisson model differ?** It models counts and imposes a log conditional mean; functional-form sensitivity is itself a limitation.
4. **What does the confidence interval mean?** Values from moderately negative to moderately positive remain compatible with the model and sampling uncertainty.
5. **What did you personally do?** Answer only after independently rerunning, verifying, rewriting, and mastering the package; do not represent AI-supported work as independently completed until that is true.
6. **Why did the timing audit matter?** The frozen June `t-1` snapshot falls inside the BDS March `t-1` to March `t` transition. The clean-preperiod model uses June `t-2`; its sign flips but remains imprecise.